Reality Seed: MIT Solve 2027 Global Challenges and the Applicant's Context (Batam, Indonesia)

Compiled: 3 November 2026.

Purpose: a fact base for independent analysis. It contains no recommendations, rankings or conclusions. Facts come from the cited sources, quoted as those sources reported them.

Labels used:

- [CONFLICT]: sources disagree.
- [UNVERIFIED]: could not be confirmed from a second source.
- [APPLICANT]: stated by the applicant.

Collection limits: facts were gathered through web search result summaries. Direct access to solve.mit.edu pages was not available, so some full documents were not read end to end.

Coverage note: some topics have more detail than others. The difference reflects which topics were searched earlier. It does **not** indicate importance, priority or likely success. Treat topics with little detail as under-researched, not as weaker.

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1. The applicant

- [APPLICANT] Lives in Batam, Riau Islands Province (Kepulauan Riau), Indonesia.
- [APPLICANT] Is a high-school graduate taking a gap year.
- [APPLICANT] Is applying alone. No team members, organisation or institutional affiliation were mentioned.
- [APPLICANT] Has not personally identified a specific problem from lived experience ("nothing comes to mind right now").
- [APPLICANT] Local communication runs mostly through WhatsApp and casual chat. In the applicant's experience, emails are often not treated as important, and there are not many bodies that are clearly responsible or responsive. The applicant contrasts this with large organisations elsewhere, such as companies whose customer support reliably replies.
- [APPLICANT] Willing to show a prototype to anyone reachable. Reach is through personal networks and WhatsApp.
- [APPLICANT] Can build software prototypes with AI coding assistance.
- [APPLICANT] Time available: about 3 days before the deadline (as of 3 Nov 2026).
- [APPLICANT] States these priorities:
  - The solution must address an actual, existing problem in society, not merely be a "helper".
  - The solution must actually solve the problem.
  - Prior validation or reach is important. The applicant notes that selected teams such as OkHi had already reached the parties who use their solution.
  - The prototype must be real and working.
  - Intended beneficiaries must actually be able to access and want to use the solution.

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2. MIT Solve: programme facts

2.1 What Solve is

- MIT Solve is an initiative of the Massachusetts Institute of Technology. It runs open innovation challenges and selects "Solver teams" each year.
- Selected teams receive funding and join a support programme.
- Winners are announced at Solve Challenge Finals in New York during the UN General Assembly and Climate Week (September).

2.2 The 2027 Global Challenges: dates

- Applications opened 22 September 2026, 9:00 am EDT.
- Deadline [CONFLICT]:
  - Solve's own challenge pages (Learning, Climate, Economic Prosperity) and the 2027 application-questions PDF state **6 November 2026, 12:00 pm EST** (noon, not end of day).
  - A Solve application-clinic event page (26 Oct) and several third-party listings state **2 November 2026, 12:00 pm EST**.
  - One outlier listing states 31 January 2027.
  - The live application portal is the authority.
- Late submissions are not accepted for any reason, including technical issues, time-zone confusion or connectivity problems.
- Judging runs in several rounds from November 2026 to March 2027. Finalist pitches take place in February 2027.
- If selected, teams commit about 25 hours over nine months, plus about four days at Solver Orientation in Cambridge, MA, in April.

2.3 The four 2027 challenges and their dimensions (as reported)

Health. "How can we use technology to make good health and quality care more equitable for all?"

1. Strengthen continuity of care in the face of outbreaks, climate shocks, conflicts or financing disruptions. Examples given: early-warning systems, health data interoperability, reliable supply chains, local production.
2. Prevention and management of chronic and later-life conditions: noncommunicable diseases, mental health conditions, healthcare needs of ageing populations.
3. Primary care access: support frontline health workers, or reduce the cost, distance and administrative barriers to care.

Climate. Asks how to adapt equitably to a warming climate while cutting emissions from how people live and eat.

1. Low-carbon, climate-resilient cities.
2. Protection from heat, floods, drought and wildfire.
3. Climate-resilient agrifood systems.
4. Ocean and coastal systems.

Learning. "How can we close global education gaps while improving outcomes for every learner?"

- The challenge page names improving assessment of what learners truly know and can do, including formative assessment, diagnostics and early screening that flag off-track learners for teachers.
- Solve wants to see gains in what learners can actually do, not just engagement or completion.
- [UNVERIFIED] The full list of Learning dimensions was not retrieved.

Economic Prosperity. Asks how to widen access to decent work, affordable housing and financial stability.

1. Financial health and resilience for individuals, families and small businesses.
2. Pathways to quality employment, especially for young people.
3. Safe, affordable, climate-resilient housing.

Indigenous Communities Fellowship. A separate track that accepts concept-stage solutions.

2.4 Eligibility

- Solutions must be technology-based. All responses and uploads must be in English.
- Global Challenges consider Prototype, Pilot, Growth and Scale stages [CONFLICT]:
  - Official pages say concept-stage solutions go only to the Indigenous Communities Fellowship.
  - Solve's 10th-anniversary materials and the application questions mention "broader eligibility for early-stage teams" and acceptance "across stages, including early-stage".
  - Some third-party sites say concept stage is accepted.
  - Solve states that the stage selected does not affect selection chances.
- One official description defines concept stage as an idea still being researched for feasibility, including prototypes under development.
- A solution cannot be the same one Solve has previously selected.
- Solve says that if the only goal is funding, Solve is not the right programme.

2.5 Judging criteria (as reported across sources)

- The criteria are: alignment with the challenge; potential for impact; feasibility, including technical feasibility; innovation; human-centred design; scalability; and partnership potential (how Solve can help).
- One source adds evidence that the technology works.
- Some criteria are applied only from the finalist stage onward.
- Review stages: staff and community screening, then expert judge scoring, then finalist pitches.

2.6 Awards

- Each selected Solver team receives a US$10,000 grant from Solve and a nine-month support programme. The total selection-stage prize pool is reported as more than US$1 million.
- One listing reports an average of about US$40,000 per selected team including partner prizes [UNVERIFIED].
- Partner prizes announced for 2027:
  - **AI for Humanity Prize** (Patrick J. McGovern Foundation): up to US$150,000 for at least one 2027 team. One listing says US$200,000 [CONFLICT].
  - **Citizens Workforce Innovation Prize:** US$100,000, for US-based teams only.

    - **Seeding the Future Prize** (climate-positive and equitable food systems): US$150,000.
    - **Crescent Enterprises AI for Social Innovation Prize:** up to US$100,000, for a maximum of four teams.
    - **E Ink Innovation Prize:** US$100,000 (sustainability, education, cross-cultural understanding).
    - **Prince Albert II of Monaco Foundation Ocean Innovation Prize:** up to US$100,000.
- In 2025, the Johnson & Johnson Foundation gave a US$100,000 Health Innovation Award across six teams. A 2027 health-specific partner prize was not found [UNVERIFIED].

2.7 Application form (partial, from fragments of the 2027 questions PDF)

- Solution name: 30-character limit.
- The following are required, with about 200 words each:
    - the specific problem;
    - the solution;
    - why the team is applying to Solve.
- Team size and time spent on the solution: about 100 words each.
- Innovation, business model and proof of success: about 300 words each.
- Challenge-specific screening questions include:
    - "Does your solution address a health or healthcare access problem?"
    - "…an economic opportunity or financial inclusion problem?"
    - "…an educational or learning problem?"
- Solve's guide warns that generic or impersonal answers, including generically AI-generated text, can seriously hurt an application. It includes guidelines on using generative AI tools.
- Other guide advice:
    - Read the judging criteria first.
    - Apply only to truly relevant challenges.
    - Describe the specific problem within the challenge.
    - Show logical links between activities, outputs and outcomes.
    - Websites that are consistently unavailable may make an application less competitive.

2.8 Selectivity

- 2023: more than 1,500 applicants from more than 120 countries.
- 2025: 31 winners selected from more than 2,900 applications from more than 130 countries; more than US$1.5 million awarded.
- 2026 (10th Anniversary Global Challenge): 10 winners from more than 1,000 applications across 110 countries, each receiving US$100,000.

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3. Past selected teams and prize winners (as reported)

- **2017:** Ruangguru Digital Bootcamp (Indonesia). It aimed to help underserved youth who had dropped out to pass national exams and gain employable skills.
- **2020, AI for Humanity Prize:** MapSights, International Wakashan AI Consortium, Ubenwa, and Moskeet (TrakitNow).
- **2023, AI for Humanity Prize:** Synnefa, Lucrefy, Climate Health Vulnerability Mapping, and GNYPWD Accessible Connect.

- **2024, Global Health Equity (selected):**
  - MediBot.
  - AmarDoctor.
  - NiADA, non-invasive anaemia detection with AI.
  - ThriveLink was a finalist.
- **2024, Global Learning (selected):** Hunu, MakerDost, NABU Multilingual Literacy Platform, Musa, and The Tatum T1. Tilli was also listed, with its status unclear.
- **2024, Climate (named in the Solver class):** SXD and Amini (an AI platform for African environmental data).
- **2025, Global Health (selected):** Clínicas del Azúcar AI Predictive Care, Rightfit, IMPALA, Impulso Previne, Smart Scope CX, and Visilant.
  - Finalists: Elucid, The Blue Box, Mental Strength, Sanku, and Alveofit's Spirometer.
- **2025, Economic Prosperity (selected):** ABALOBI, ELEVATE, Vinco, SwarajAbility for PwD, and the OkHi Digital Addressing System.
  - Finalists: LoanReady by Cyphr, AgriBaato, Green Finance 3.0 with Aloi, PehchanPe, and Mona.
  - Semi-finalists: Haqdarshak and AAvance Wallet.
- **2025, Indigenous Communities Fellowship (selected):** Indigenize Toys, Ag-based solutions, Sisters of the Sea, Kumu Connect, and Onvkuce: Voice Database.
- **2025, E Ink Innovation Prize:** Adaptive Reader, BeeLine Reader, Housing NOW, and IMPALA.
- **2025, J&J Health Innovation Award (examples):** AI-Driven Diabetes Care (Monterrey, Mexico) and IMPALA (Netherlands).
- **2026, 10th anniversary winners (US$100,000 each):**
  - Citizens Workforce Innovation Prize: Making Space.
  - Collegium Prize: Screening Linked to Care (Rural Behavioral Health Institute).
  - E Ink Innovation Prize: Flint Paper Battery.
  - Schmidt Marine Wavemaker's Prize: BlueGea Group.
  - Seeding the Future Prize: BioIQ (Adaptive Symbiotic Technologies).
  - Solver Prize: Kara Solar.
  - AI for Humanity Prize (US$200,000 in 2026): Karya and Rocket Learning.
- **Related Indonesia-linked items:**
  - Bookbot, a reading app with Bahasa Indonesia speech recognition, was one of 10 winners of the Octava Social Innovation Challenge, which was run on the MIT Solve platform.
  - Kipin (Indonesia) was reported as a winner of an "MIT x Octava Foundation" challenge.
- **Solutions with Solve solution pages** (this does not by itself mean they were selected, because applicants also get pages):
  - Hikma Health: an offline open-source EHR for refugee clinics, used by 14 organisations in 10 countries.
  - Simprints: biometric digital ID.
  - Golden Dreams (Issara Institute).
  - Ulula.
  - SAHAT Chatbot.

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4. Batam and Riau Islands: local facts

4.1 Population and economy

- Population [CONFLICT, different definitions]:
  - BPS mid-2025 estimate: 1,296,960.
  - Population and Civil Registry (Disdukcapil) records: 1.39 million residents in the second half of 2025.
- Batam is a Free Trade Zone and Free Port (KPBPB). It was set up under Government Regulation 46/2007 and revised by Government Regulation 47/2025. BP Batam manages licensing, land, infrastructure, ports and investor facilitation.
- Manufacturing was 57.01% of Batam's gross regional product in 2025. The economy grew 6.76% in 2025, and 7.49% year on year in Q4 2025.
- 2025 investment commitments: US$10.349 billion (Rp172.3 trillion).
- Batam is about 20 km from Singapore.

4.2 Work and income

- Batam's unemployment rate is 7.57% (2025, BPS via Databoks; Disnaker head).
- Registered job seekers:
  - 21,434 as of August 2025.
  - [UNVERIFIED] 15,189 registered and 5,242 placed in January–June 2026 (Batam Pos, citing Disnaker).
- The 2026 minimum wage (UMK) for Batam is Rp5,357,982 per month.
- Layoffs:
  - December 2025: three companies filed layoff requests covering about 2,600 workers, including about 1,800 at PT Esun.
  - Mid-2026: PT Mega Solar stopped operating, and hundreds of workers were furloughed or laid off.
  - October 2026: about 145–150 workers at PT Yisheng Photovoltaic faced layoff because the company had no orders.
  - [UNVERIFIED] HKI reported 26 manufacturing companies closed.
- In Riau Islands, SMA/SMK (high school) graduates were 63.63% of the unemployed (February 2026). Nationally, SMK graduates had the highest unemployment rate: 8.63% (August 2025).
- A recruiter-published article reports that Batam factories struggle to find mechatronics and automation skills.
- Unions criticised the 2026 outsourcing regulation (Permenaker 7/2026). They say some shipyard outsourcing firms pay below the minimum wage.

4.3 Recruitment fraud and recruitment fees

- **2025 Batam case:** two women posing as HR assistants of PT Sumitomo (Batamindo) took Rp700,000–1,000,000 per person as a "formality" fee. Victim counts and amounts:
  - Batam Pos and Kompas: 140 victims, about Rp140 million in total.
  - Batamnews: "hundreds" of victims, losses of hundreds of millions of rupiah.
  - Police said the scheme ran from January 2025. A suspected organiser is on the wanted list.

- **Other Batam cases:** a Polsek Batam Kota case (February 2025) and a Batam District Court case (10 victims, Rp16.1 million).
- **Scam methods reported in Batam:**
  - Fake HR WhatsApp accounts that use company logos.
  - APK file links sent via WhatsApp or Telegram as "job info", which are phishing.
  - "Like and rate products" task scams.
- **Broker fees:**
  - Batam: brokers (calo) asked Rp3–6 million, as reported by the FSPMI Batam chair (date uncertain, around 2022).
  - Java: Rp3–15 million in Karawang (Deputy Minister of Manpower), and Rp4–7 million or more in Cikarang.
  - Older reporting describes a pattern where applicants are told the factory has no openings, but are placed through an outsourcing foundation (yayasan) after paying, reportedly around Rp5 million.
  - A 2020 Transparentem investigation found Indonesian apparel workers paid middlemen one to six months' wages to secure real jobs, and sometimes paid again at contract renewal.
- **Banten cases:** in October 2026, 10 job seekers paid Rp13–33 million each for a promised permanent factory job. In a 2025 Tangerang case, the perpetrator claimed closeness to the company.
- **Legal position:**
  - Law 13/2003 Article 38 restricts placement fees.
  - Permenaker 18/2024 bars private placement agencies from charging job seekers placement fees.
  - Kepmen 230/2003 allows fees for certain positions (e.g. manager and supervisor levels), capped at one month's wage.
- **Enforcement:** Serang regency formed an anti-extortion task force on factory brokers (2025–2026). West Java's governor targeted HR staff who sell jobs (October 2026).
- **National data:**
  - OJK's Indonesia Anti-Scam Centre (IASC) logged 18,220 job-fraud reports with Rp656 billion in losses from November 2024 to 15 October 2025. Job fraud was the 4th of 10 scam types.
  - The job-fraud count reached 23,469 by 14 January 2026.
- **Survey:** the GASA 2025 survey (1,000 adults) found 49% had experienced a job scam (denominator unclear) and that 89% of scam attempts arrived via WhatsApp.
- **Platforms and government:**
  - Komdigi took down 7,908 fake overseas job adverts in January–August 2026.
  - SEEK reports about 1% of Jobstreet Indonesia ads are fake, and that Indonesia accounts for about 38% of its Asia-Pacific scam attempts.
- Many recruitment-related Batam companies state publicly that their recruitment is free.

4.4 Migration and trafficking

- Batam is a transit point for undocumented ("non-procedural") migrant workers going to Malaysia and Singapore. Main routes are the Batam Centre ferry terminal and back routes near the Barelang bridge.

- Batam immigration postponed the departure of 974 passengers suspected of non-procedural migration in October 2026 (804 of them at Batam Centre), and 1,009 in September 2026.
- Police uncovered three trafficking networks in July–August 2026. Recruits were financed to travel to Batam, then sent to Malaysia.
- BP3MI Kepri reported that more than 5,000 Riau Islanders had been trafficked to Cambodia (February 2025).
- Indonesians seeking embassy help to leave Cambodian scam operations:
  - 12,019 in the first half of 2026.
  - 5,088 in all of 2025.
  - The Foreign Ministry counted 6,800 Indonesians involved in online-scam cases by October 2025.
- Officials state that Cambodia is not a legal placement country for Indonesian migrant workers.
- Returnees often face suspicion at home. Some return to scam compounds voluntarily.

4.5 Water, power and environment

- **Water crisis (2026):**
  - Residents of Nongsa district reported no piped water for 3–6 months.
  - They blocked access to Nongsa Digital Park on 5 October 2026, one of several protests in two months.
  - Residents link the shortages to data-centre construction. BP Batam denies this and cites drought and low flows at the Nongsa reservoir.
  - Tanker trucks were sent, and supply was restored to about 53% of affected areas within days.
- A former mayor estimated Batam's clean water can support about 1 million residents.
- A blackout on 12 October 2026 hit Batam and Bintan. Backup generators could run only about 30% of normal water production.
- Mangroves: a BPDAS (forestry agency) note says Batam lost 40% of its mangroves in 10 years. Bengkong fishers report falling catches linked to reclamation (Mongabay, December 2025).
- Batam has 103 flood-prone zones (May 2025). BP Batam leaders blamed mismanaged land and drainage.

4.6 Social

- Orang Suku Laut (sea-nomad) children are often out of school and lack ID documents. Some leave school by junior high because of bullying.
- Batam had 6,554 children out of school (2024).
- Batam had more than 42,000 informal homes (ruli) in 2014. Tembesi residents were evicted for an industrial estate (January 2025).
- In March 2026, Batam authorities reported rising mental-health incidents. No counts were given.

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5. Indonesia: national facts by challenge area

5.1 Digital access

- Internet penetration was 80.66% in 2025 (APJII; 229.4 million users of 284.4 million people; survey of 8,700 respondents aged 13+). 83.39% of internet users go online via smartphone. Java accounts for 58% of users.
- Internet use in 2025: urban 82.72%, rural 70.14% (BPS).
- Mobile phone ownership (BPS, 2025):
  - 93.50% of households own or control a mobile phone.
  - 70.75% of individuals do.
  - In 2023 the figures were 89.02% of households and 67.29% of individuals.
- Some districts are much lower. In Nusa Tenggara Timur, about 41% of people aged 5+ in Sumba Barat own a phone, and about 62% in Belu.
- A national WhatsApp user count was not found.

5.2 Health

- **Tuberculosis:** Indonesia has the 2nd-highest burden in the world, about 10% of global cases (WHO Global TB Report 2025). The 2024 report cited about 1.06 million cases and 134,000 deaths a year.
- **Stunting** among children under 5 (SSGI 2024):
  - 19.8% nationally, down from 21.5% in 2023 (about 4.48 million children).
  - 29.8% in the poorest quintile.
  - Targets: 18.8% in 2025 and 14.2% by 2029.
- **Hypertension** (SKI 2023): 30.8% of adults 18+ had it by measurement, but only 8.6% reported a prior diagnosis.
- **Diabetes** (SKI 2023): 11.7% of people had above-normal blood sugar.
- **Depression** (SKI 2023): prevalence is highest at ages 15–24, and only 10.4% of young people with depression sought treatment.

5.3 Climate and disasters

- BNPB recorded about 3,116–3,176 disasters in 2025 [CONFLICT, by cutoff date]. About 99% were hydrometeorological.
- Floods: 1,454–1,584 events (sources differ). Extreme weather: 638–673 events.
- Hydrometeorological disasters have made up more than 97% of events every year since 2015.

5.4 Learning

- **PISA 2022:**
  - More than 3 in 4 Indonesian 15-year-olds did not reach minimum proficiency (Level 2) in mathematics and reading.
  - Reported scores: reading about 355–359, mathematics about 359–366, science 376 [CONFLICT on exact values].
  - The gap between socio-economically advantaged and disadvantaged students in mathematics was 34 points. The OECD average gap is 93 points.

- **Learning trend (2000–2014):** a study of Indonesian learning over 14 years (Beatty et al., International Journal of Educational Development 2021; RISE) found:
  - learning fell by about a quarter of a standard deviation, while enrolment rose;
  - the average Grade 7 student in 2014 had the numeracy of the average Grade 4 student in 2000;
  - the decline affected all subgroups and was not explained by changes in who was enrolled.
- **INOVASI study** of 18,370 pupils in grades 1–3: about half of grade 3 pupils were below minimum literacy, and two in three were below minimum numeracy.
- **Assessments:**
  - The national assessment (Asesmen Nasional) tests a sample of grade 5 pupils and gives no individual scores.
  - The new grade 6 academic test (TKA) results are accessed only through school operators.
- The Education Minister has said 75% of Indonesian children can read but lack comprehension.
- Pemantik, an assessment toolkit for teachers modelled on India's ASER, sorts literacy into 4 levels and numeracy into 5. Its authors note that many teachers follow the textbook without tracking each child's progress.
- In April 2026, the Ministry of Primary and Secondary Education partnered with the Tanoto Foundation, the Gates Foundation and UNICEF on foundational literacy and numeracy, including structured pedagogy with real-time diagnostic assessment.
- Globally, an estimated 70% of 10-year-olds cannot understand a simple written text (2022 joint report; 57% before the pandemic).

5.5 Economic prosperity and finance

- **IASC (OJK):**
  - 702,342 scam reports from November 2024 to 30 September 2026.
  - Rp811.1 billion of victims' funds blocked; Rp207.16 billion returned.
  - Reported losses of Rp9.1 trillion by 14 January 2026 (reported, not verified).
- **Illegal online lending (pinjol ilegal):**
  - OJK's task force (Satgas PASTI) shut down 2,263 illegal lenders in 2025.
  - It received 22,632 complaints about illegal lenders between January 2025 and 14 January 2026, 3,403 of them from Sumatra.
  - OJK runs a WhatsApp legality check on 081-157-157-157.
- **Kemnaker (Ministry of Manpower):**
  - Karirhub listed 81,171 active vacancies from 32,086 companies (September 2026).
  - Employers must report vacancies to Kemnaker, with enforcement from 2026 (Perpres 57/2023).
  - A QR code per vacancy has been announced to fight fake jobs. Its launch status is unknown.
  - Kemnaker has a fake-vacancy task force and complaint post.
- **KP2MI SISKOP2MI:** a public portal listing licensed overseas placement agencies (P3MI) and about 289,000 official overseas vacancies.

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6. Global facts relevant to the 2027 dimensions

6.1 Health continuity

- **After the 2025 US funding cuts:**
  - An amfAR survey (166 organisations, 46 countries) found more than 1,700 HIV treatment sites closed.
  - Terminated USAID awards supported about 2.3 million people on HIV treatment (CGD Note 391).
  - 77,163 fewer children received PEPFAR-supported treatment in FY2025 (−14.2%).
- **Uganda and Zimbabwe:** an interrupted time-series study (Lancet Global Health 2026) found HIV services fell by half to two-thirds after stop-work orders and stayed down through March 2026.
- **Stop-work orders took effect immediately.** For example, at 16 Ugandan regional referral hospitals, more than 60,000 patients were at risk.
- **Clinic survey, 38 LMICs, late 2025** (medRxiv preprint), disruptions reported by clinics:
  - community activities: 68%;
  - tracing of patients lost to follow-up: 67%;
  - adherence support: 63%.
- UNAIDS reported clinics without staff, and HIV medicine stock-outs in DR Congo, Ethiopia and Kenya.
- UNAIDS modelling projects 6.6 million additional HIV infections and 4.2 million AIDS deaths by 2029 if the cuts are permanent.
- **Disasters:**
  - Pakistan's 2022 floods damaged or destroyed about 2,000 health facilities (about 10% of the national total). WHO reported disrupted TB and HIV treatment.
  - After Cyclone Idai (Mozambique, 2019), continuity of HIV care broke down. One Beira health centre saw advanced HIV cases rise about 25%.
- After a flood in Japan, medication interruption among outpatients with chronic conditions was 9% overall and 23% among evacuees.
- A systematic review found 18 studies naming medication continuity as a key priority in disasters.
- **Displacement:** in conflict settings (Sudan, Gaza), records were lost and responsibility for chronic care shifted from institutions to individuals.
- **WHO guidance:** people established on HIV treatment should be offered 3–6-month refills (6 preferred), subject to supply and storage.
- **Noncommunicable diseases** killed at least 43 million people in 2021 (75% of non-pandemic deaths). 82% of premature NCD deaths occur in LMICs (WHO, September 2025).
- **Early warning:** 119 countries (60%) report multi-hazard early-warning systems. Disaster mortality is nearly 6 times lower where those systems are comprehensive (WMO 2025).

6.2 Selected intervention evidence (by domain)

Learning

- Targeting instruction to a child's learning level (Teaching at the Right Level) is rated among the most cost-effective education interventions.

- **ConnectEd (Youth Impact):** weekly SMS problems plus a weekly 20-minute phone tutoring call for grades 3–5, run in about 4,500 households in Botswana.
  - Learning gains were found in every location tested across six countries, averaging about 0.30–0.35 standard deviations.
  - Cost: about US$12 per child.
  - It works on basic phones.
- **Parents' beliefs (Malawi, Dizon-Ross):** parents' beliefs about their children's performance were inaccurate, and giving them accurate information changed how they invested in their children's education.
- **India WhatsApp parent study:** parents in a WhatsApp arm became more accurate at choosing the correct next learning step.

Jobs and fraud

- **Indonesian migrant workers (Bazzi et al., RCT):** information on recruiter quality reduced use of low-quality recruiters and improved outcomes, although beliefs did not change; workers searched longer instead.
- **US mail fraud trial:** warning letters reduced repeat victimisation by 8.6% (one letter) and 22.4% (repeated mailings) over four months.
- **Scam-detection training:** gains decay within 2–3 weeks.
- No published evaluation of losses prevented was found for Singapore's ScamShield bot.

Health and SMS

- In a US Medicare pilot, tailored interactive text messages raised refill rates by 14.07 percentage points.
- A Haiti pilot of SMS reminders for NCD patients found it feasible where patients have readable phones.
- An Egyptian diabetes SMS trial found improved adherence.

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7. Existing solutions and incumbents (landscape, as found)

Recruitment and jobs (Indonesia)

- Kemnaker Karirhub / SIAPkerja, with mandatory vacancy reporting, a planned QR code and a complaint post.
- KP2MI SISKOP2MI, the overseas agency and job-order registry.
- BP3MI Denpasar's "Bali Mantap" app for checking licensed agencies (Bali only).
- Job platforms with fraud teams: Jobstreet, Glints (TapLoker) and KitaLulus (10M+ users; takes reports via WhatsApp).
- ITUC/ILO Recruitment Advisor: worker reviews of agencies, available in Bahasa Indonesia, with 1,152 Indonesian reviews.

Scams and fact-checking

- Mafindo's Kalimasada (2020) and CekSumber (2025, AI): WhatsApp chatbots for hoax and scam checking.
- CekRekening.id and AduanNomor: account and number blacklists.
- OJK IASC: reporting and freezing of funds.
- Comparable tools elsewhere: Singapore ScamShield Bot (forward-and-check, police blacklist plus machine learning), Malaysia Semak Mule (37M+ visits), and the Philippines DMW

licensed-agency lookup.
- Truecaller for Business.

Finance

- OJK's WhatsApp legality check for lenders.

Learning

- Bookbot: a reading app for children with Bahasa Indonesia speech recognition, supported by INOVASI and UNICEF.
- Rocket Learning: a WhatsApp programme for parents of young children in India. It states that it does not know how each child is doing.
- Wadhwani AI: automated oral reading-fluency assessment (India).
- ConnectEd / Youth Impact: phone tutoring in Botswana and other countries; no Indonesian deployment found.
- Pemantik: an assessment toolkit for teachers.
- Ruangguru: a commercial edtech platform in Indonesia.

Health

- Hikma Health: an offline EHR for refugee clinics.
- Simprints: biometric ID.
- Medic / Community Health Toolkit, Dimagi CommCare and OpenSRP: platforms for community health workers.
- Simple.org (Resolve to Save Lives): hypertension management.
- Google Flood Hub and GloFAS: flood forecasts.
- Red Cross Early Action Protocols: pre-agreed funding for anticipatory action.

Climate and coast

- Global Mangrove Watch and Nusantara Atlas: mangrove and forest monitoring.
- SEWA / Climate Resilience for All: parametric heat insurance for informal women workers (India).

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8. Technical facts relevant to prototyping

- **WhatsApp Cloud API (Meta):**
  - The free test phone number can message only up to 5 pre-registered recipient numbers.
  - Temporary access tokens expire in 24 hours.
  - Using your own number requires registering and verifying it. Display-name review has no fixed timeline.
  - Business verification unlocks higher messaging limits.
  - Messages are charged per conversation. Current 2026 rates for user-initiated conversations were not verified.
- **Open-Meteo:** free, keyless weather-forecast and flood (GloFAS-based river discharge) APIs.
- **Smartphones:** 83.39% of Indonesian internet users go online via smartphone. Basic phones support voice calls and SMS.

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9. Data caveats and corrections

- The deadline conflicts across sources (6 Nov vs 2 Nov). Check the portal.
- The Sumitomo-impersonation case is reported as 140 victims (Batam Pos, Kompas) and as "hundreds" (Batamnews).
- These figures could not be independently confirmed: 15,189 / 5,242 job seekers in January–June 2026; the "26 manufacturing companies closed" figure.
- Batam's population figure depends on the source (BPS estimate vs civil registry).
- Recruitment fees are not illegal for every role. Kepmen 230/2003 permits capped fees for some positions.
- Having a Solve "solution page" does not mean the team was selected.
- The full wording of the 2027 Learning challenge dimensions was not retrieved.
- Several facts come from news reports and search-result summaries rather than full primary documents.

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